



JASHORE UNIVERSITY OF SCIENCE AND TECHNOLOGY

Lab Report on,

SQL Commands for Table Manipulation

Course: Database Management System Lab

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Lab Report: SQL Commands for Table Manipulation

Experiment Name:

Using SQL Queries for Table Creation and Manipulation

Introduction:

In this experiment, we explore basic SQL commands essential for database management. We cover creating tables, retrieving specific data using SELECT and WHERE, inserting data, modifying the table structure by adding new columns, and deleting tables. These fundamental operations are critical for effective data organization, storage, and retrieval in any relational database.

Methodology:

The experiment involves the following SQL commands:

1. **Creating a Table (CREATE TABLE):** This command defines a new table structure.

```
CREATE TABLE Students (  
  StudentID INT PRIMARY KEY,  
  FirstName VARCHAR(50),  
  LastName VARCHAR(50),  
  Age INT  
);
```

Here, a Students table is created with columns for StudentID, FirstName, LastName, and Age.

After running this commend

StudentID	FirstName	LastName	Age
(INT)	(VARCHAR 50)	(VARCHAR 50)	(INT)

2.Inserting Data into a Table (INSERT INTO): This command adds records to the table.

```
INSERT INTO Students (StudentID, FirstName, LastName, Age)
VALUES (1, 'John', 'Doe', 21);
```

This inserts a record into the Students table with specified values.

After running this commend

StudentID	FirstName	LastName	Age
1	John	Doe	21

3.Selecting Data (SELECT and WHERE): These commands retrieve specific records.

```
SELECT FirstName, LastName
FROM Students
WHERE Age > 20;
```

This retrieves the FirstName and LastName of students who are older than 20.

After running this commend

FirstName	LastName
John	Doe

4.Adding a New Column (ALTER TABLE ADD COLUMN): This command modifies the table structure.

```
ALTER TABLE Students
ADD COLUMN Major VARCHAR(50);
```

Here, a new column Major is added to the Students table.

After adding a column, the table structure now includes `Major`, with `NULL` values for existing records:

StudentID	FirstName	LastName	Age	Major
1	John	Doe	21	NULL

5.Deleting a Table (DROP TABLE): This command removes the table from the database.

DROP TABLE Students;

1. This permanently deletes the Students table.
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Result & Discussion:

Each SQL command successfully performed its respective function:

1. **Table Creation:** The Students table was created with specified columns, establishing the structure for data storage.
 2. **Data Insertion:** The INSERT INTO command added data accurately, confirming the table's readiness to store and manage student information.
 3. **Data Selection:** The SELECT and WHERE commands efficiently retrieved only the records that met the conditions, validating the effectiveness of conditional queries.
 4. **Column Addition:** Adding the Major column displayed SQL's flexibility in modifying table structures without losing data.
 5. **Table Deletion:** The DROP TABLE command removed the table, indicating how SQL enables flexible database management.
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Conclusion:

The experiment demonstrated the core SQL commands essential for table management. These commands form the foundation of database operations, enabling users to create, modify, retrieve, and delete data structures as needed. Mastery of these commands is fundamental for database administration and manipulation.